

PRODUCT REQUIREMENTS

KamiCode PRD

AI-native competitive programming, developer skill graphs, and
verified credentials.

AI-Native Evaluation

Developer Skill Graph

Verified Credentials

Startup Build Plan

Product Requirements Document

KamiCode is an AI-native competitive programming platform that evaluates code beyond correctness, builds a developer skill graph, and turns meaningful achievements into portable verified credentials.

1. Product Positioning

Positioning: a verified developer skill graph for students, developers, competitive programmers, colleges, and recruiters.

Tagline: Prove how well you code, not just how much you code.

Strategic decision: The product is skill-first and AI-powered. Blockchain is a credential layer, not the main product.

Developer Value

Practice, compete, receive AI feedback, build a public skill profile, and prove achievement context.

Recruiter Value

Verify candidate skills through topic mastery, speed, consistency, complexity quality, and public credentials.

College Value

Track student placement readiness, run contests, and generate objective skill reports.

Differentiator

Combines deterministic judging with AI reasoning, skill graph analytics, and portable achievement verification.

2. Problem Statement

- Existing coding platforms mostly show solved count, badges, or rating without deeper performance context.
- A recruiter cannot easily verify whether a candidate writes efficient code consistently across topics.
- Students lack a strong public profile that captures growth, speed, topic mastery, and contest performance.
- Traditional badges are centralized and not easily portable across hiring or education ecosystems.

3. Product Goals

- Create a daily coding habit through competitive challenges and streaks.
- Use AI to analyze complexity, approach, optimization quality, and explain improvements.
- Build a public developer skill graph that can support placement and hiring.
- Create verified credentials for only meaningful achievements such as top ranks, streaks, and contest wins.
- Start with a focused MVP and evolve into a startup-grade competitive learning platform.

4. Target Users

Persona	Need	KamiCode Promise
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Engineering Student	Placement readiness, proof of DSA skill, motivation.	AI feedback, skill graph, public profile, verified progress.
Competitive Programmer	Ratings, leaderboards, speed, hard problem proof.	Daily rankings, puzzle rush, contest history, achievement credentials.
Self-Taught Developer	Credibility without traditional degree signals.	Portable proof of practical coding performance.
Recruiter	Trustworthy candidate skill evidence.	Topic-wise profile, verified achievements, performance summaries.

5. Core Product Modules

Module	Description
Problem System	Problem bank, daily challenge, topic tags, hidden test cases, editorials, hints.
Submission Engine	Code editor, language selector, sandbox execution, runtime and memory metrics.
AI Evaluation	Time complexity, space complexity, approach detection, optimization advice, quality score.
Skill Graph	Topic mastery, speed, accuracy, consistency, growth, public profile.
Competitive Layer	Leaderboards, daily streaks, puzzle rush, contests, ratings, divisions, seasons.
Credential Layer	Internal badges first, then optional NFT or soulbound credentials for major milestones.

6. MVP Scope

The first serious version should prove one complete loop: solve problem -> run tests -> AI analysis -> score -> profile update -> achievement.

- One daily problem
- One primary language at launch: Python
- Monaco code editor
- Docker-based code execution
- AI complexity and feedback result
- Daily leaderboard
- Public user profile
- Internal achievement badges
- Optional credential minting after core loop is stable

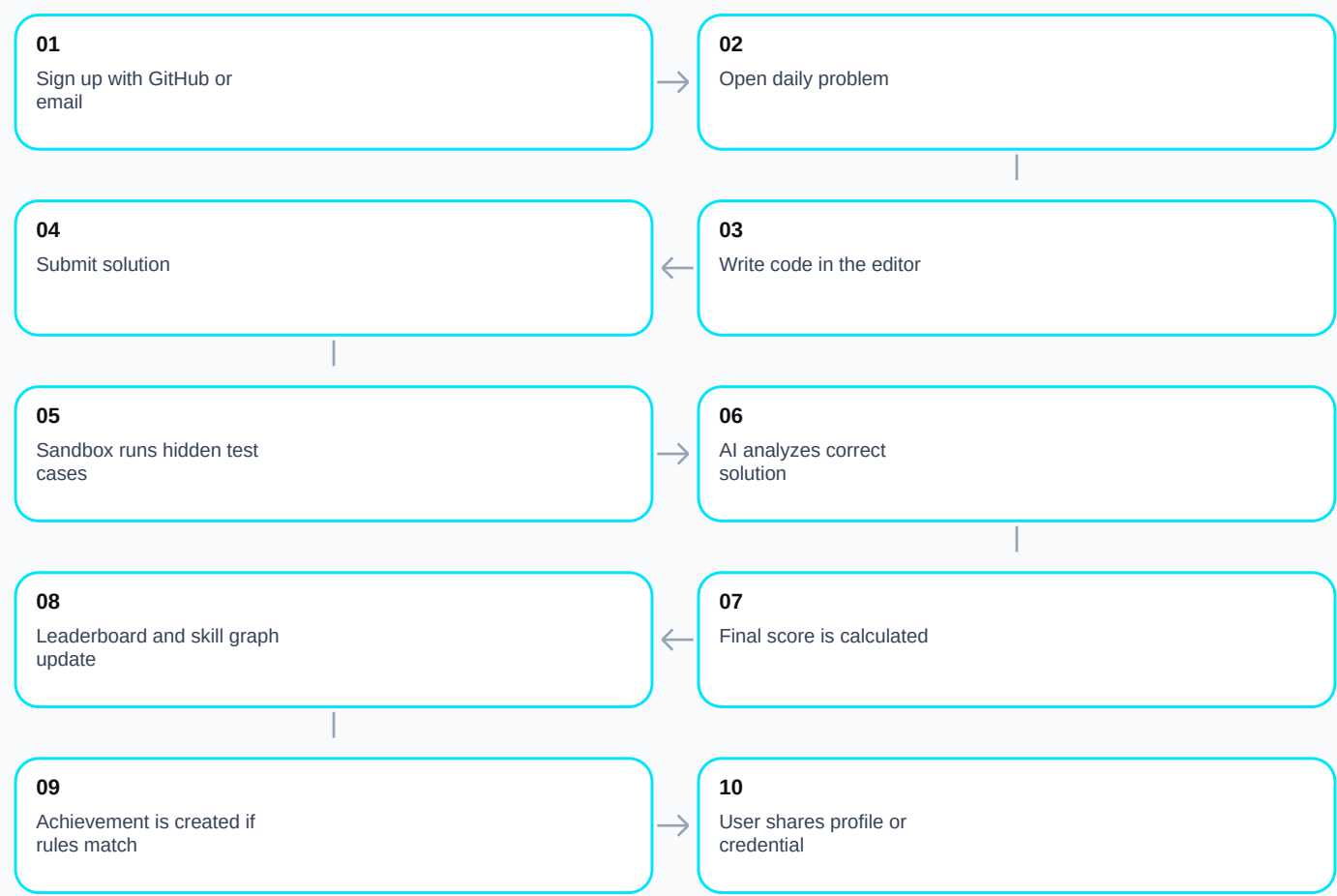
7. Feature Roadmap

Phase	Focus	Features
V1	Core Coding Platform	Auth, problem page, code editor, deterministic judging, result page, basic leaderboard
V2	AI Differentiation	Complexity detection, approach explanation, optimization suggestions, code quality score
V3	Skill Graph	Topic analytics, strengths/weaknesses, progress dashboard, public share card
V4	Competition	Streaks, weekly contests, puzzle rush, ratings, divisions, seasonal rankings
V5	Credentials	Verified pages, NFT/soulbound credentials, public verification links

V6	Hiring Layer	Recruiter search, college dashboards, candidate coding reports
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8. User Journey

User Product Flow



9. Requirements

Category	Requirement
Functional	Auth, problem browsing, code submission, test execution, AI analysis, scoring, leaderboard, profile, achievements.
Non-Functional	Fast feedback, secure sandboxing, scalable execution workers, reliable storage, clear observability.
Trust Rules	Correctness must be deterministic. AI should explain and evaluate quality, not decide pass/fail.
Security	No network access in code containers, CPU/memory limits, rate limits, input sanitization, server-side credential validation.

10. Success Metrics

- Daily active solvers
- Submission completion rate

- AI feedback satisfaction
- Profile share rate
- Weekly retention
- Contest participation
- Credential verification views
- Recruiter or college pilot interest

11. Startup Summary

KamiCode should become a developer reputation platform, not just a coding website. The long-term business opportunity is in trusted skill verification for students, colleges, communities, and hiring teams.